

The Complete Injection–Suction and First-Cusp Atlas for Quadratic Polubarinova–Galin Flow

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Abstract

For the normalized quadratic map $f(\zeta, t) = a(t)\zeta + b(t)\zeta^2$ we solve the constant-rate Polubarinova–Galin equation on its entire smooth univalent branch. The complex coefficient a^2b is conserved, normalized area is affine in physical time, injection is global and asymptotically circular, while noncircular suction ends at an explicitly timed ordinary semicubical cusp. Circular collapse and nonsmooth or invalid initial faces are separated. This round is the **coefficient-reduction owner**.

1 Normalized flow and exact reduction

Let $a(t) > 0$, $b(t) \in \mathbb{C}$, and

$$f(\zeta, t) = a(t)\zeta + b(t)\zeta^2, \quad \operatorname{Re}[f_t \overline{\zeta f_\zeta}] = q \quad (|\zeta| = 1), \quad (1)$$

where $q \in \mathbb{R}$ is constant. The normalization $f(0, t) = 0$ and $f_\zeta(0, t) = a(t) > 0$ fixes translation and rotation. A *smooth univalent solution* means $a > 2|b|$; then f is conformal on a neighborhood of $\overline{\mathbb{D}}$ and parametrizes a smooth Jordan boundary.

Theorem 1 (Complete smooth coefficient branch). *For smooth univalent initial data, equation (1) is equivalent to*

$$\dot{a} = \frac{aq}{a^2 - 4|b|^2}, \quad \dot{b} = -\frac{2qb}{a^2 - 4|b|^2}. \quad (2)$$

On its unique maximal smooth interval,

$$\kappa := a^2b \quad \text{is constant}, \quad M_0 := a^2 + 2|b|^2 = \frac{\operatorname{Area}(f(\mathbb{D}))}{\pi}, \quad \dot{M}_0 = 2q. \quad (3)$$

Put $u = a^2$ and $u_c = (4|\kappa|^2)^{1/3}$, taking $u_c = 0$ when $\kappa = 0$. Then

$$M_0 = F(u) := u + \frac{2|\kappa|^2}{u^2}, \quad F'(u) = 1 - \frac{4|\kappa|^2}{u^3}, \quad (4)$$

and smooth univalence is exactly $u > u_c$.

If $q > 0$, the solution is smooth for every $t \geq 0$, u increases to infinity, and $2|b|/a \rightarrow 0$. If $q = 0$, the map is stationary.

Proof. On $|\zeta| = 1$, expanding the left side of (1) and equating its constant and first Fourier modes gives

$$a\dot{a} + 2\operatorname{Re}(\dot{b}\bar{b}) = q, \quad a\dot{b} + 2\dot{a}b = 0. \quad (5)$$

The determinant is $a^2 - 4|b|^2$, so (2) follows. The second identity gives $(a^2b)^\cdot = 0$, and both identities give $\dot{M}_0 = 2q$. The area formula follows either from the coefficient area theorem or from $\frac{1}{2} \int_0^{2\pi} \operatorname{Im}(\bar{f}f_\theta) d\theta$. Substituting $b = \kappa/u$ proves (4).

Indeed,

$$f(z_1) - f(z_2) = (z_1 - z_2)\{a + b(z_1 + z_2)\}, \quad f'(\zeta) = a + 2b\zeta.$$

Thus $a > 2|b|$ gives injectivity and nonvanishing derivative on $\overline{\mathbb{D}}$; equality puts the sole critical point on its boundary, and $a < 2|b|$ puts it inside. Since F is strictly increasing on (u_c, ∞) , $M_0(t) = M_0(0) + 2qt$ uniquely determines that branch. For $q > 0$ it exists globally with $u \rightarrow \infty$ and $2|b|/a = 2|\kappa|/u^{3/2} \rightarrow 0$. For $q = 0$, (2) vanishes. $\square$

2 Sharp suction endpoint and cusp normal form

This revision is the **first-cusp owner**.

Theorem 2 (First singularity under suction). *Assume $q < 0$ and smooth univalent initial data. If $\kappa \neq 0$, the maximal smooth time is*

$$T = \frac{M_0(0) - M_c}{-2q}, \quad M_c = F(u_c) = \frac{3}{2}u_c. \quad (6)$$

As $t \uparrow T$, $u \downarrow u_c$, $a \downarrow \sqrt{u_c}$, and $|b| \uparrow \sqrt{u_c}/2$. No self-intersection on $\mathbb{D}$ and no critical point in $\mathbb{D}$ occurs earlier. At T the limiting boundary has exactly one ordinary semicubical cusp at

$$\zeta_c = -\frac{a_c}{2b_c}, \quad z_c = f(\zeta_c, T) = -\overline{b_c}. \quad (7)$$

Proof. On the upper branch $u > u_c$, equation (4) is strictly increasing and its infimum is $F(u_c) = 3u_c/2$. The affine area clock therefore reaches that value at exactly (6); before then the strict quadratic criterion in the preceding proof excludes both critical points and self-intersections. At the endpoint write $b_c = Be^{i\phi}$, so $a_c = 2B$, and set $\zeta = -e^{-i\phi}e^{is}$. Then

$$\begin{aligned} e^{i\phi}\{f(\zeta, T) - z_c\} &= B\{-2e^{is} + e^{2is} + 1\} \\ &= -Bs^2 - iBs^3 + \frac{7}{12}Bs^4 + O(s^5). \end{aligned} \quad (8)$$

With $X = -\operatorname{Re}(e^{i\phi}(z - z_c))$ and $Y = -\operatorname{Im}(e^{i\phi}(z - z_c))$, one has $X = Bs^2 + O(s^4)$, $Y = Bs^3 + O(s^5)$, and

$$\lim_{s \rightarrow 0} \frac{Y^2}{X^3} = \frac{1}{B}. \quad (9)$$

This is the ordinary semicubical normal form and proves uniqueness of the boundary cusp. □